



ST BENEDICT'S

SUBJECT	PHYSICAL SCIENCE: CHEMISTRY
GRADE	11
EXAMINER	Mr McCoy
MODERATOR	Mr Hilder
DATE	25 July 2023
MARKS	165
DURATION	2,5 hours

%

LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of **28** pages. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae, below, whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Additional space is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8
Mark	GR	ZN	SG	41-4.5: ZN 15	KH	FM	KH	FM
Moderated Mark				4.6:SG 16				
Question Total	16	21	24	31	8	21	27	17

Comment:

QUESTION 1 GR**8 MARKS**

1.1 D✓✓

1.2 C✓✓

1.3 C✓✓

1.4 D✓✓

1.5 C✓✓

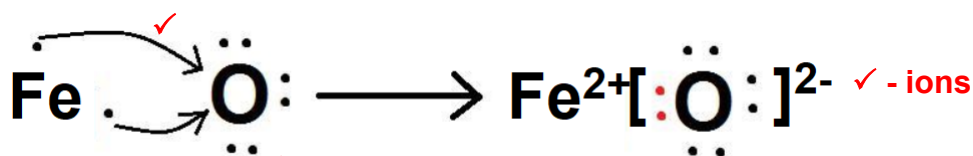
1.6 D✓✓

1.7 D✓✓

1.8 B✓✓

QUESTION 2 CHEMICAL BONDING**21 MARKS**

- 2.1 $2\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$ balanced (4)
- 2.2 Explain, in terms of its bonding and structure, why iron (III) oxide...
- 2.2.1 Ionic bonding; many strong electrostatic forces of attraction that need a lot of energy to be overcome. (4)
- 2.2.2 Ions are fixed in position. (1)
- 2.3 Iron has metallic bonding. Fixed lattice of positive kernels and a sea of electrons that are free to move. (4)
- 2.4 Iron (III) oxide has a melting point of 1565°C while iron (II) oxide has a melting point of 1377°C .



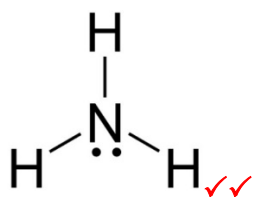
- 2.4.1 **Lewis diagrams** (3)
- 2.4.2 A transferring of electrons between metal and non-metal. (2)
- 2.4.3 The electrostatic attraction in Iron (III) oxide is greater than Iron (II) oxide. Iron (III) oxide has a cation with a 3+ charge, where as Iron (II) oxide has a cation with a 2+ charge. The 3+ change will attract the anion/ O^{2-} with a greater force. (3)

QUESTION 3 CHEMICAL BONDING**24 MARKS**

- 3.1 ATOMS✓ (1)
- 3.2 Electron pairs are shared✓ between two non-metal atoms✓ (2)
- 3.3 Electrons are equally shared✓ because there is zero difference in electronegativity✓ (2)
- 3.4 Ammonia is a polar molecule.

3.4.1 Electronegativity difference (END) = $3 - 2,1 = 0,9$ ✓
 END is greater than zero✓
 therefore bonds are polar✓ (3)

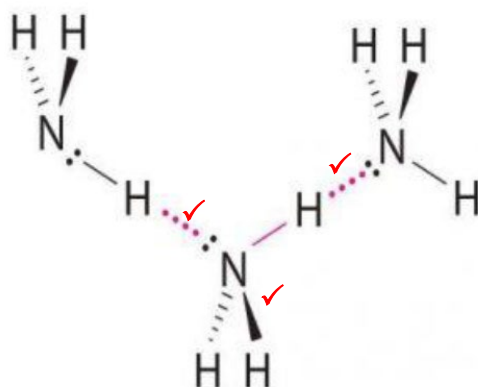
3.4.2 The molecule does NOT have electron cloud symmetry✓ due to its pyramidal shape. ✓



(4)

3.4.3 Ammonia liquefies at $-33,1\text{ }^{\circ}\text{C}$.

a) Hydrogen bonding✓. (1)



b) (3)

3.5 Nitrogen gas liquefies at $-195,8\text{ }^{\circ}\text{C}$.

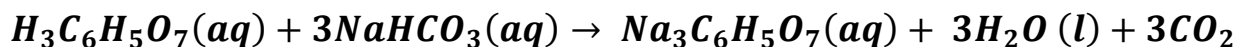
3.5.1 London forces (1)

3.5.2 As the nitrogen molecules approach each other (get close) ✓
 there is a dispersion of electrons in each nitrogen molecule✓
 resulting in induced/ temporary dipoles forming✓
 which are then able to attract each other✓ (4)

3.6 liquid ammonia will have hydrogen bonding between its molecules (coe 3.4.3a)
 liquid nitrogen will have London force between its molecules (coe 3.5.1)
 hydrogen bonding is stronger than London force✓
 hence, ammonia will liquify first✓✓ (3)

QUESTION 4 ENERGY AND RATES**15 MARKS**

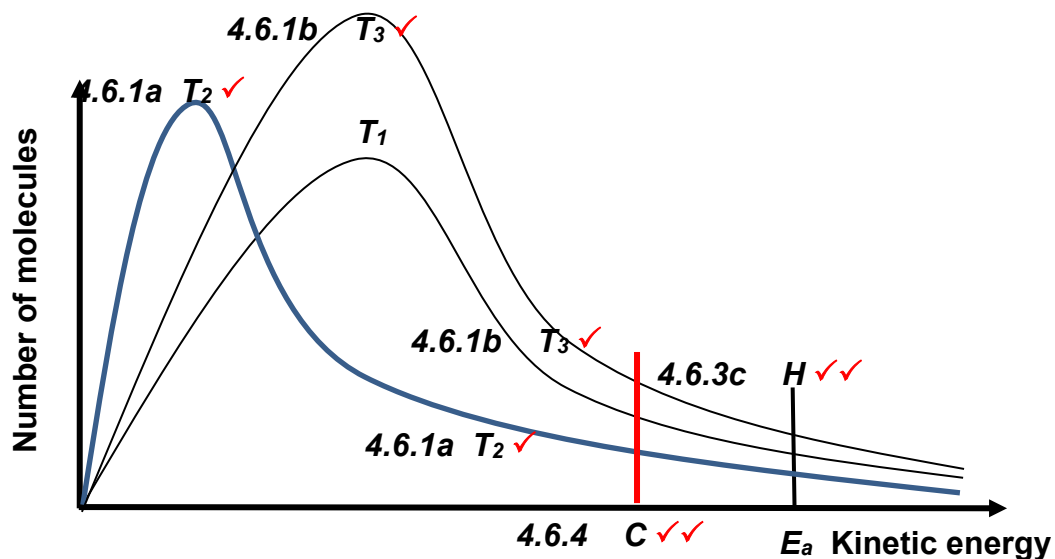
The reaction of citric acid and sodium bicarbonate produces sodium citrate and water in addition to the carbon dioxide, according to the following balanced chemical equation:



Consider the heat of reaction.

- 4.1 the net change of chemical potential energy✓ of the system✓ (2)
- 4.2 as reactions which transform thermal energy✓ into chemical potential energy✓ (2)
- 4.3 ENDOTHERMIC✓ (1)
- 4.4 ENDOTHERMIC✓ (1)
- 4.5 .
- 4.5.1 The loss of mass✓
carbon dioxide is a heavy gas✓
which can escape the vessel✓ (3)
- 4.5.2 So that the mass lost is only due to the escaping gas✓
as any liquids trying to escape✓ due to splashing will be trapped✓ (3)
- 4.5.3 A timer (1)
- 4.5.4 A inverted measuring cylinder – downward displacement of water✓
Volume of gas would be measured✓
OR a gas syringe. (2)

- 4.6 The graph below shows the Maxwell-Boltzmann distribution curve for the reaction and a given temperature, T_1 . The activation energy for the reaction, E_a , is marked.



- 4.6.1 On the graph above: (6)
- 4.6.2 the minimum energy ✓ required to start a chemical reaction ✓ OR the energy required to form the activated complex (2)
- 4.6.3 substance that increases the rate of the reaction ✓ but remains unchanged at the end of the reaction ✓ (2)
- 4.6.4 On the graph: (2)
- 4.6.5 Since there are more particles with $E_k \geq E_a$ ✓✓, there will be more effective (OR successful) collisions ✓ per unit time ✓. (4)

QUESTION 5 SOLUTIONS**8 MARKS**

5.2 Beaker A: $n = 0,75 \times 0,25 = 0,1875 \text{ mol} \times 3 = 0,5625 \text{ mol}$ of NH_4^+

Beaker B: $n = 20 / (14 + 4 + 14 + 3(16)) = 0,25 \text{ mol}$

$$c = \frac{0,5625 + 0,25}{0,25 + 0,5}$$

$$c = \frac{0,8125}{0,75}$$

$$c = 1,08 \text{ mol} \cdot \text{dm}^{-3}$$
 (4)

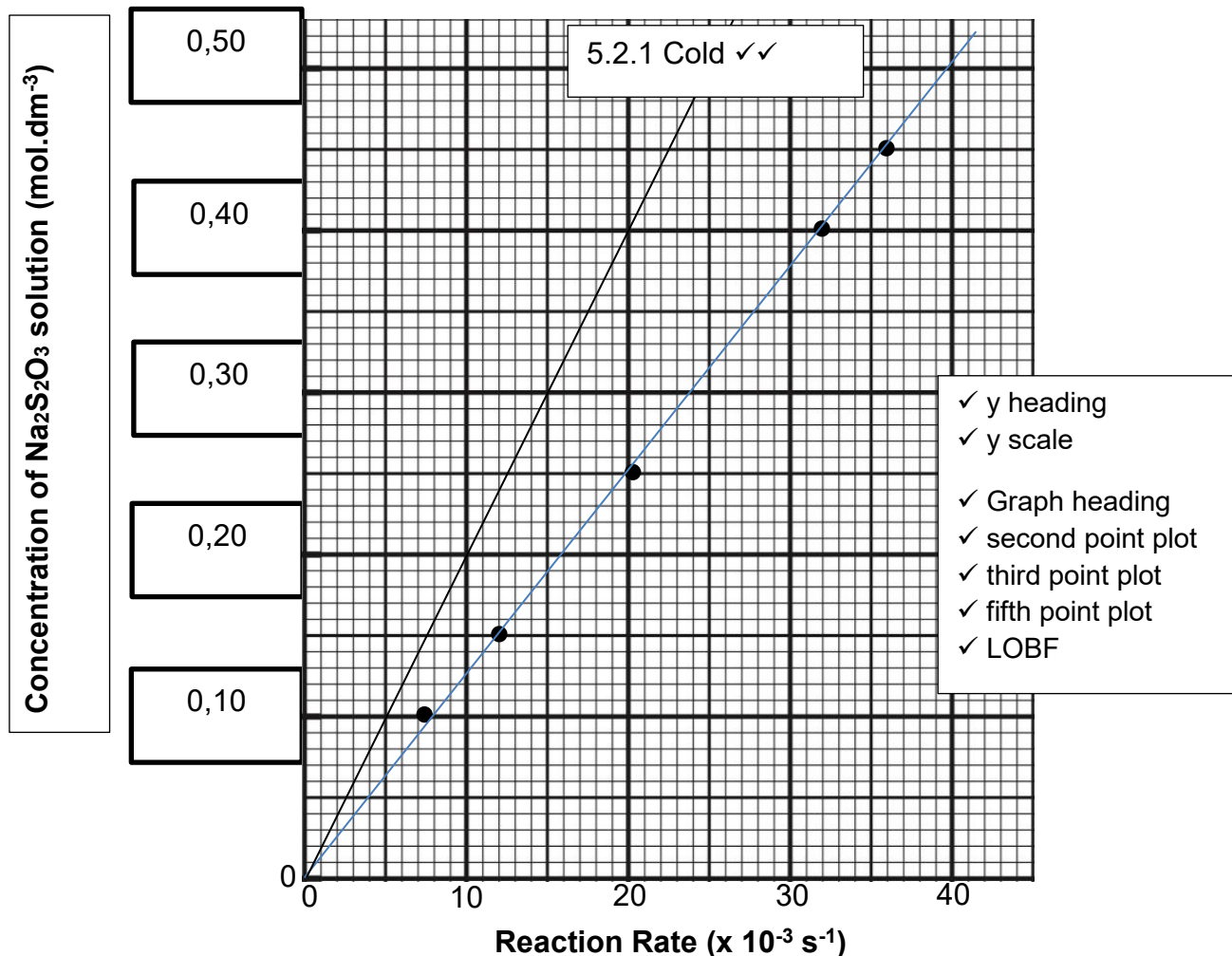
QUESTION 6 ENERGY AND RATES**21 MARKS**

6.1 .

6.1.1 25✓,0✓ (s) (2)

6.1.2 Reaction rate = Average⁻¹✓✓ (2)

6.1.3 Graph of concentration vs reaction rate✓ (7)



6.2 .

6.2.1 ON GRAPH (2)

6.2.2 Decreased kinetic energy. ✓
 The number of collision will decrease. ✓
 There will be fewer effective collisions✓ (with sufficient energy and correct orientation) per second. ✓ Hence, the reaction rate will decrease (4)

6.3 .

6.3.1 (Priyesh would be looking through a greater volume of solution) there would be more of the precipitate forming✓ in his line of sight to the cross✓, therefore making the cross disappear quicker.
 NOT ACCEPTING - Priyesh would be looking through a greater volume of solution (2)

6.3.2 10 (s) ✓✓ (2)

QUESTION 7 KH

7.1 Al_2O_3 ✓✓ (2)

7.2 .

7.2.1 NaOH ; $n = cv$

$$= 2 \times 0,2$$

$$= 0,4 \text{ moles available } \checkmark$$

H_2O : $n = m/M$

$$= 12/18$$

$$= 0,67 \text{ moles available } \checkmark$$

NaOH : H_2O

$$2 : 3$$

$$0,4 : 0,6 \text{ Enough } \checkmark$$

0,44: 0,67 Not enough

0,67 moles of H_2O available, but only 0,6 required. Therefore NaOH is limiting ✓ (4)

7.2.2 measure of the extent of a reaction ✓ generally measured by comparing the amount of product against the amount of product that is possible ✓ (2)

7.2.3 Theoretical yield of $\text{NaAl}(\text{OH})_4$: NaOH : $\text{NaAl}(\text{OH})_4$

$$1:1$$

0,4 moles of $\text{NaAl}(\text{OH})_4$ produced ✓

Mass of $\text{NaAl}(\text{OH})_4$: $m = n \times M$ ✓

$$= 0,4 \times 118$$

$$= 47,2 \checkmark$$

$$\% \text{ yield} = \text{actual} / \text{theo} \times 100$$

$$= 45/47.2 \times 100 \checkmark$$

$$= \mathbf{95,33\%} \checkmark$$

(5)

7.3 .

7.3.1 Fe: $n = m/M$

$= 1000\checkmark / 56$

$= 17,86 \text{ mols produced } \checkmark$



2: 1 $\checkmark$

8,93 moles of Fe_2O_3 required (3)

7.3.2 $N_{\text{Fe}_2\text{O}_3} = n \times N_A$



$= 8,93 \times 6,02 \times 10^{23} \checkmark$

2: 1

$= 5,38 \times 10^{24} \text{ Iron oxide units } \checkmark$

$2 \times 5,38 \times 10^{24} \checkmark = \mathbf{1,075 \times 10^{25} \text{ Fe atoms } \checkmark}$

(4)

7.3.3 moles of CO: 26,79 moles $\checkmark$

$V = nV_m$

$= 26,79 \times 22,4 \checkmark$

$= 600 \text{ dm}^3 \checkmark$ unit and value

(3)

7.3.4 original mass of Fe_2O_3 required : $m = nM$

$= 8,93 \times 160 \checkmark$

$= 1428,8 \text{ g } \checkmark$

Mass x 70% = 1428,8 $\checkmark$

Mass = 2041,14 g of Fe_2O_3 required $\checkmark$

OR $8,93 / 0,7 = 12,75$
moles of
 Fe_2O_3

Mass: $m = nM$

$= 12,75 \times 160$

$= 2041,14 \text{ g of } \text{Fe}_2\text{O}_3$

QUESTION 8 FM

8.1 water that is associated with crystal formation ✓✓

OR . "water of crystallization" refers to water that is found in the crystalline framework of a salt. ✓ (2)

8.2 number of water molecules ✓ in the crystal per unit of the salt ✓ (2)

8.3 Standard solution ✓ (1)

8.4

$N = cV$ $= 0,2 \times 10$ $= 2 \text{ mol} \checkmark$ $V = n/c$ $= 2/2 \checkmark$ $= 1 \text{ dm}^3 - 200 \text{ cm}^3 = 800 \text{ cm}^3 \checkmark$	$c_1v_1 = c_2v_2 \checkmark$ $10 \times 200 = 2 \times v_2$ $V_2 = 1000 \text{ ml} \checkmark$ Therefore, 800 ml must be added (0.8 dm³) ✓
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(3)

8.5 HCl: $n = cv$

$$= 2 \times 0,0235$$

$= 0,047$ moles of HCl reacted ✓ **0,0235 moles of Na₂CO₃ reacts ✓ (3)**

Na₂CO₃ : HCl

1: 2 ✓

8.6 Mass of water released: $5 - 2,491 = 2,509 \text{ g} \checkmark$

Moles of water = m/M

$$= 2,509/18 \checkmark$$

= 0,139 moles ✓ (3)

8.7 Ratio of water to dehydrated sodium carbonate H₂O : Na₂CO₃

$$0,139 : 0,0235 \checkmark$$

$$5.93: 1 \checkmark$$

Therefore x = 6 ✓ (MUST be a whole number) (3)